Exp 18 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
N = 1000;
sps = 20;
SNR = 20;

% Generate Random Binary Data
data = randi([0 1],1,N);
symbols = 2*data - 1;

% Upsampling
tx = zeros(1,N*sps);
tx(1:sps:end) = symbols;

% Pulse Shaping Filter
h = ones(1,sps);
signal = conv(tx,h,'same');

% Introduce Inter-Symbol Interference
isi_filter = [1 0.5 0.25];
isi_signal = filter(isi_filter,1,signal);

% Add AWGN Noise
signal_power = mean(isi_signal.^2);
```

```
noise_power = signal_power/(10^(SNR/10));
noise = sqrt(noise_power)*randn(size(isi_signal));
rx = isi_signal + noise;

% Eye Diagram Without ISI
figure;
eyediagram(signal(1:2000),2*sps);
title('Eye Diagram Without ISI');

% Eye Diagram With ISI
figure;
eyediagram(rx(1:2000),2*sps);
title('Eye Diagram With ISI and AWGN');

% Calculate BER
sample_points = sps:sps:length(rx);
received_symbols = rx(sample_points);
detected_data = received_symbols > 0;

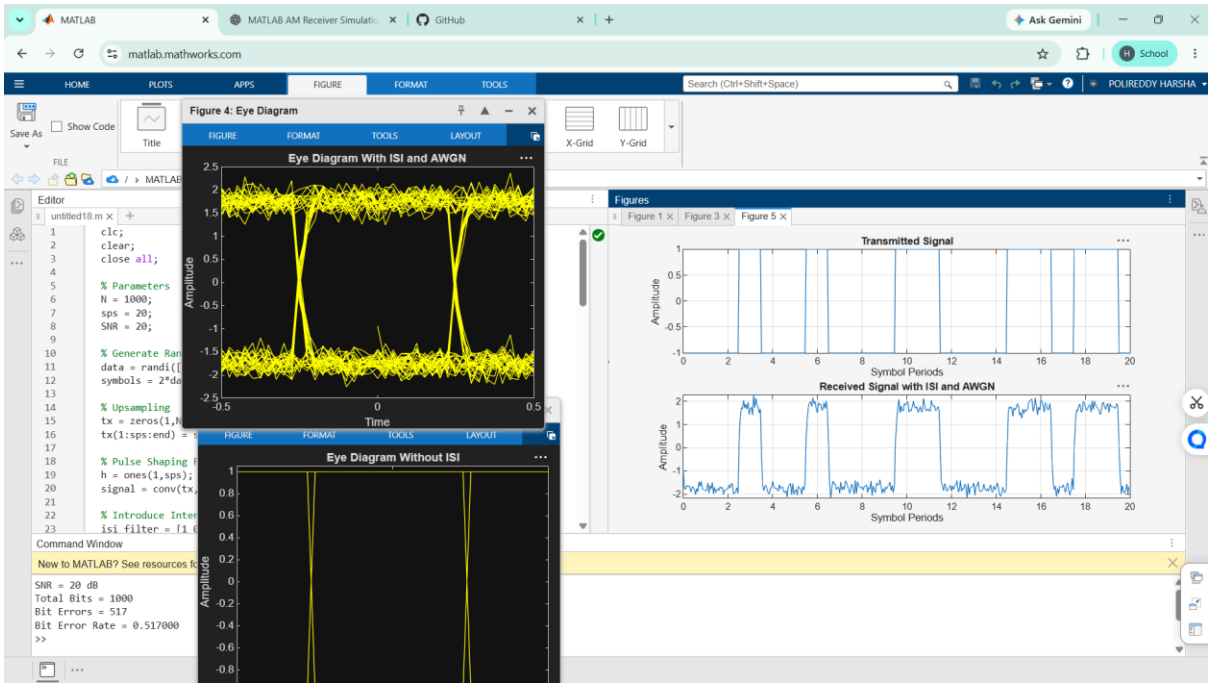
L = min(length(data),length(detected_data));
errors = sum(data(1:L) ~= detected_data(1:L));
BER = errors/L;

% Display Results
fprintf('SNR = %d dB\n',SNR);
fprintf('Total Bits = %d\n',L);
fprintf('Bit Errors = %d\n',errors);
fprintf('Bit Error Rate = %.6f\n',BER);
```

```
% Plot Transmitted and Received Signals
t = (0:length(signal)-1)/sps;

figure;
subplot(2,1,1);
plot(t(1:400),signal(1:400));
title('Transmitted Signal');
xlabel('Symbol Periods');
ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(t(1:400),rx(1:400));
title('Received Signal with ISI and AWGN');
xlabel('Symbol Periods');
ylabel('Amplitude');
grid on;
```


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Figure 4: Eye Diagram
FIGURE
FORMAT
TOOLS
LAYOUT
Eye Diagram With ISI and AWGN
Amplitude
Time
Editor
untitled18.m
clc;
clear;
close all;
% Parameters
N = 1000;
sps = 20;
SNR = 20;
% Generate Ran
data = randi([
symbols = 2*da
% Upsampling
tx = zeros(1,N
tx(1:sps:end) =
% Pulse Shaping F
h = ones(1,sps);
signal = conv(tx,
% Introduce Inter
isi_filter = [1 0
Eye Diagram Without ISI
Figures
Figure 1
Figure 3
Figure 5
Transmitted Signal
Amplitude
Symbol Periods
Received Signal with ISI and AWGN
Amplitude
Symbol Periods
Command Window
New to MATLAB? See resources fo
SNR = 20 dB
Total Bits = 1000
Bit Errors = 517
Bit Error Rate = 0.517000
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